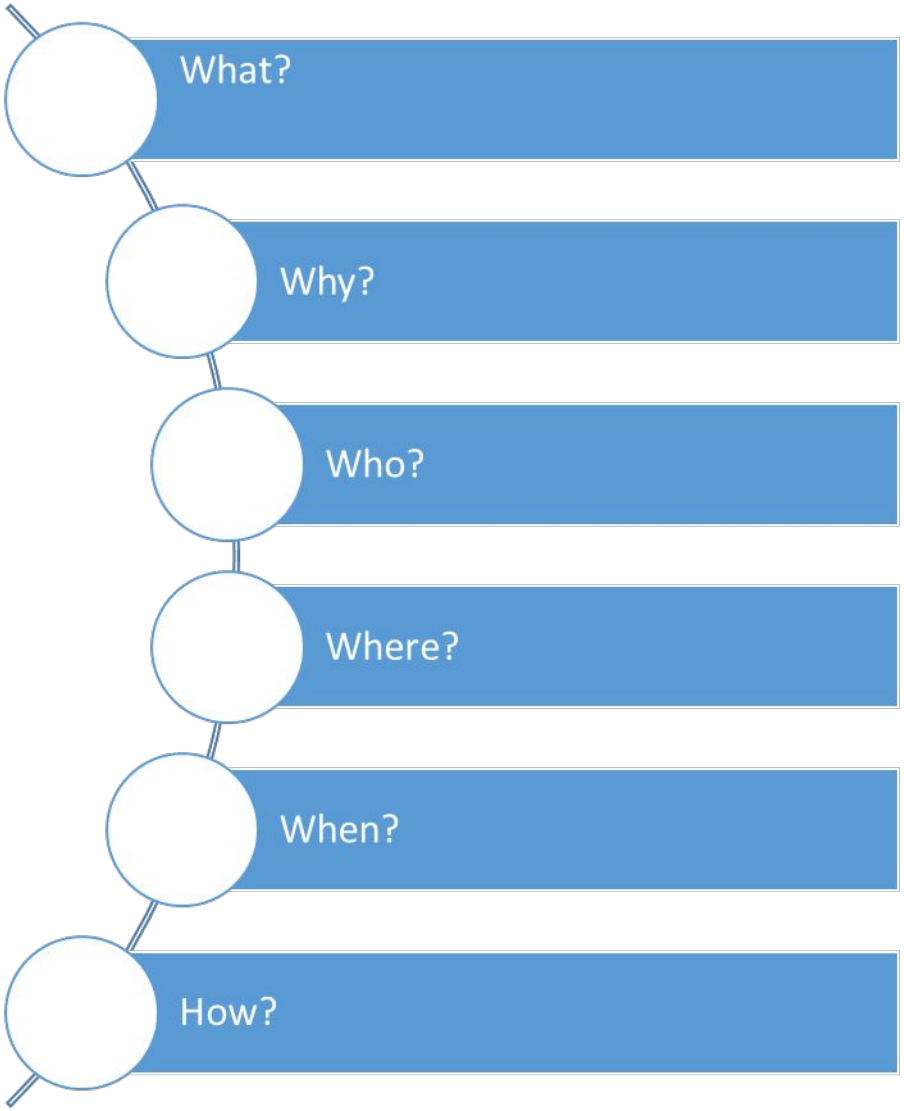
Helping Children Learn

Meta-cognition and use of talk

Mediate learning

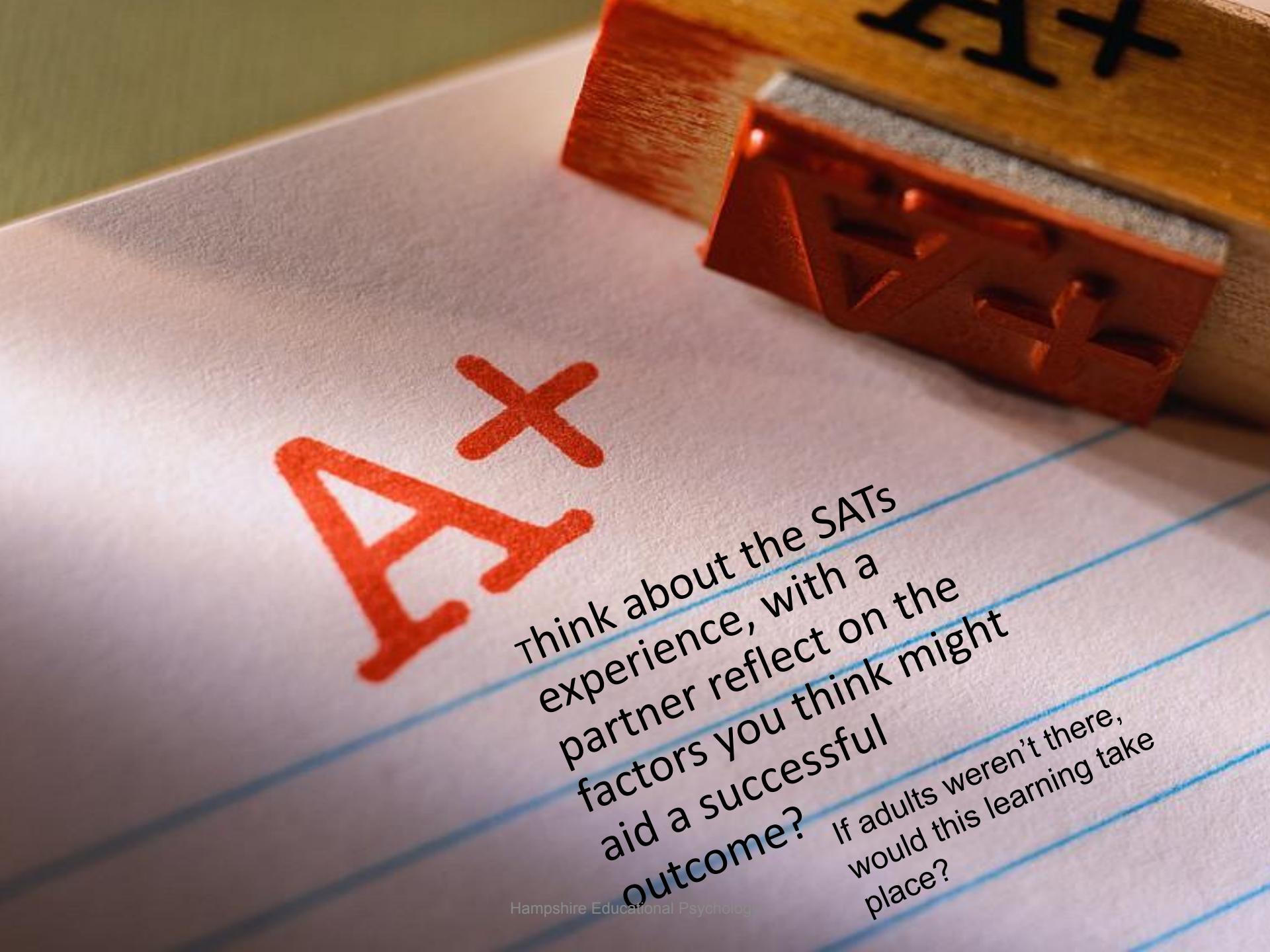
Lucy Manger
Beth Turner
Rebecca Blyth

Outline



The six oranges puzzle

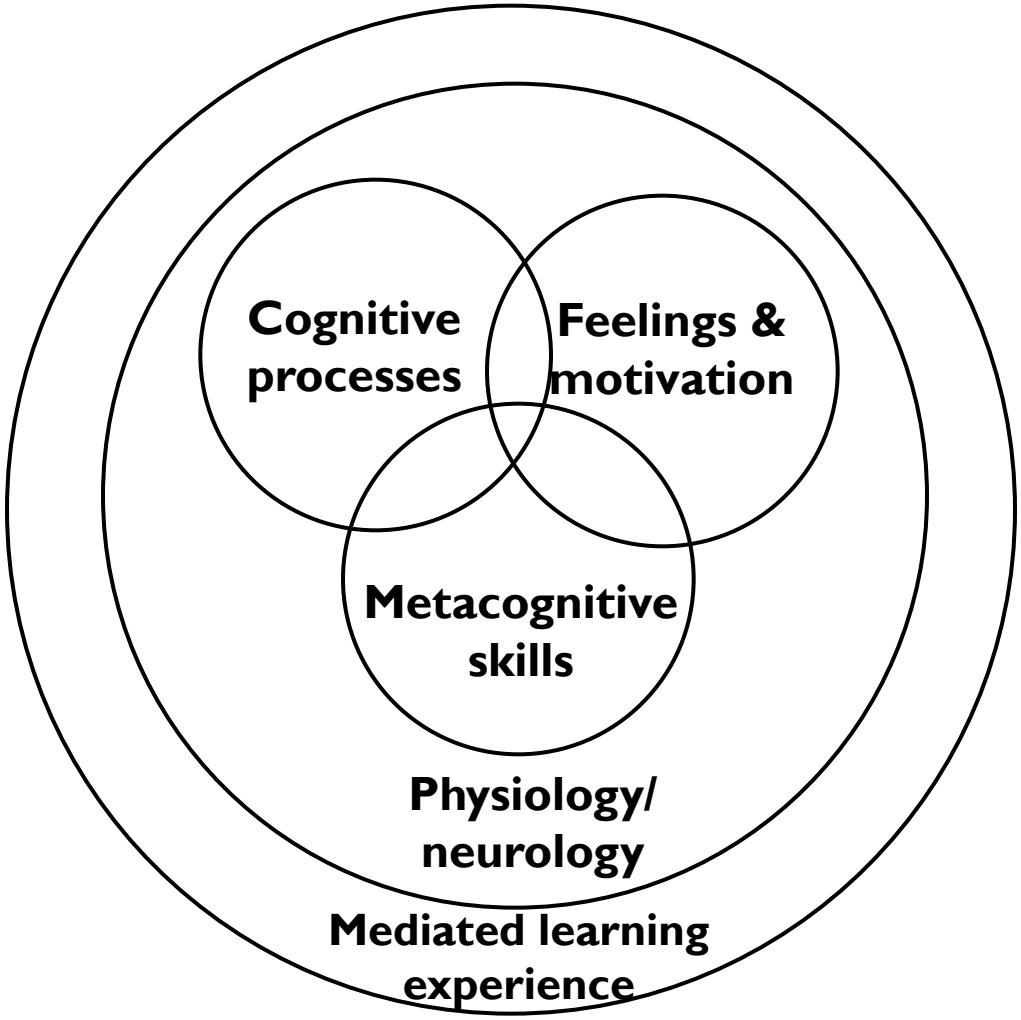
There are six oranges in the bag. How can you give these six children an orange each and still have one left in the bag



Think about the SATs experience, with a partner reflect on the factors you think might aid a successful outcome?

If adults weren't there, would this learning take place?

How we learn


Cognitive processes
Feelings & motivation
Metacognitive skills
Physiology/ neurology
Mediated learning experience

Cognition

Metacognition

Feelings and motivation

Locus of control

Fear of failure/defensiveness

Confidence in correct responses

Vitality and alertness

Accessibility to mediation

Need for mastery/intrinsic motivation

Frustration tolerance

What do we mean by “mediated learning”

MLE is a special quality of interaction between an adult, a learner and task

Why should we be interested in MLE?

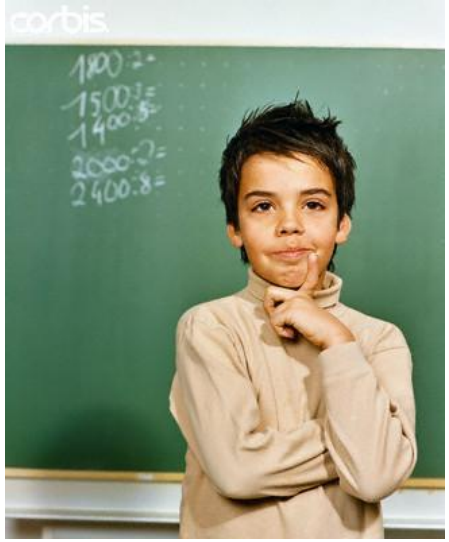
It is the cause of individual differences in cognitive development

Within child and environmental factors are secondary

MLE (or its lack) is the primary reason for learning difficulties

Who is a mediator?

How does MLE work?



Task → Learner → Response

How does MLE work?

Mediated learning

Task Adult Learner Adult Response ***Adult***

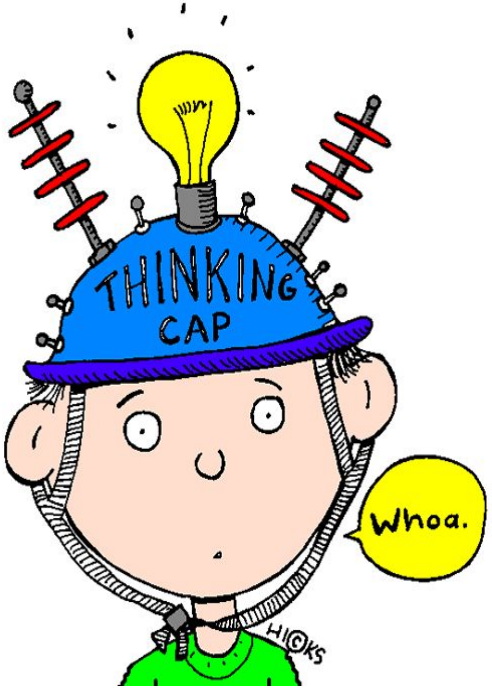
Traditional assumptions about learning	**Assumptions about learning from cognitive psychology**
Focus on content and acquiring knowledge	Focus on learning how to learn
Learning is a product	Learning is a process
Instructor	Mediator

Intelligence is:	**Cognitive processes are:**
Largely genetic (inherited)	Taught and learned
Related to a range of aptitudes (verbal, spatial, numerical)	Influenced by some fundamental competence **and** by learning and problem solving habits, motivation, attitudes and strategies
Largely fixed	Modified through mediated learning experience
Linked to the ability to learn culturally based knowledge	Generalisable across domains of knowledge

The point is, that what I can do today with your help, I can do by myself tomorrow – assuming that you have provided a MLE

Back to the six oranges puzzle

Reflecting upon learning



- What did you learn recently?
- Why did you learn it?
- How did you learn it?
- How do you know that you learnt it?

Solving a reasoning problem

Work with a few people

Can you solve this problem?

How did you solve it?

What helped you solve it?

Joe, Jack and John ran a race. Jack was not the winner. He was just a little faster than Joe. Who is the winner?

Joe

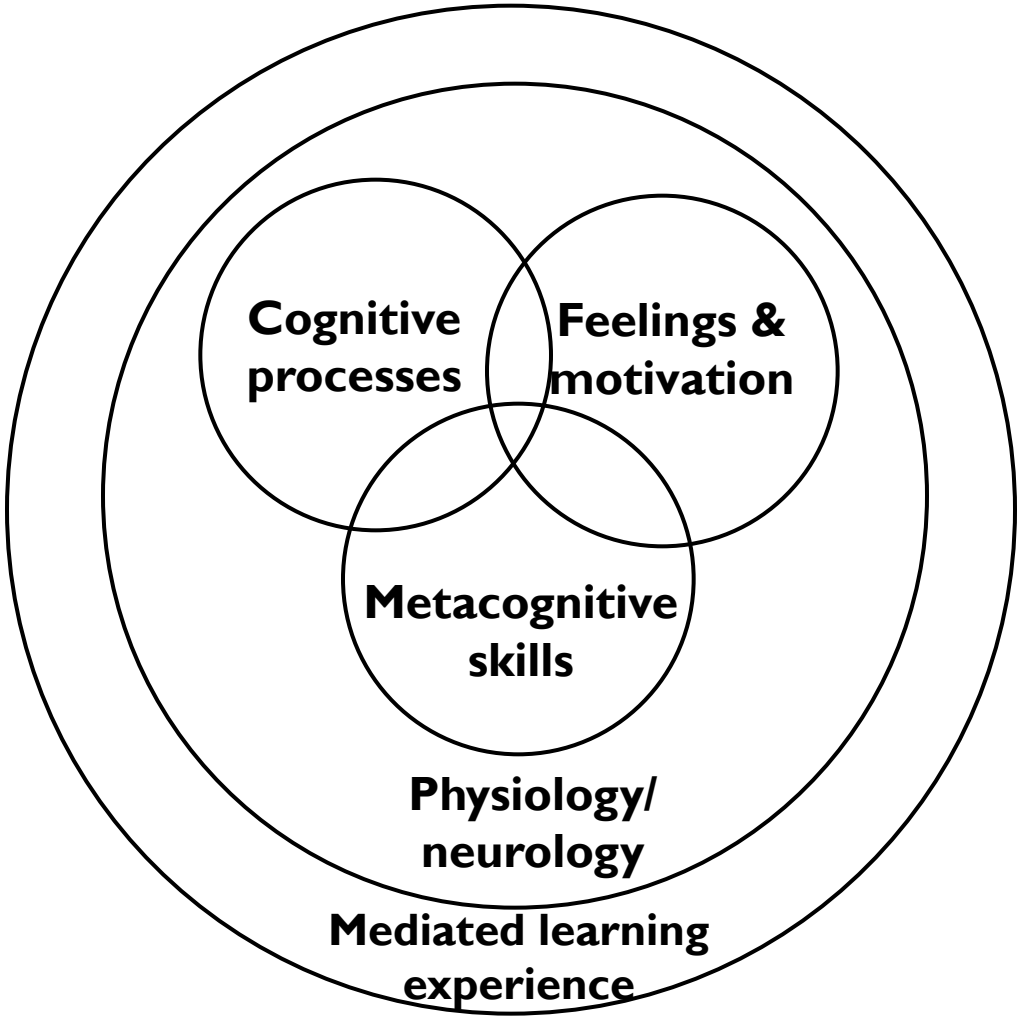
Jack

John

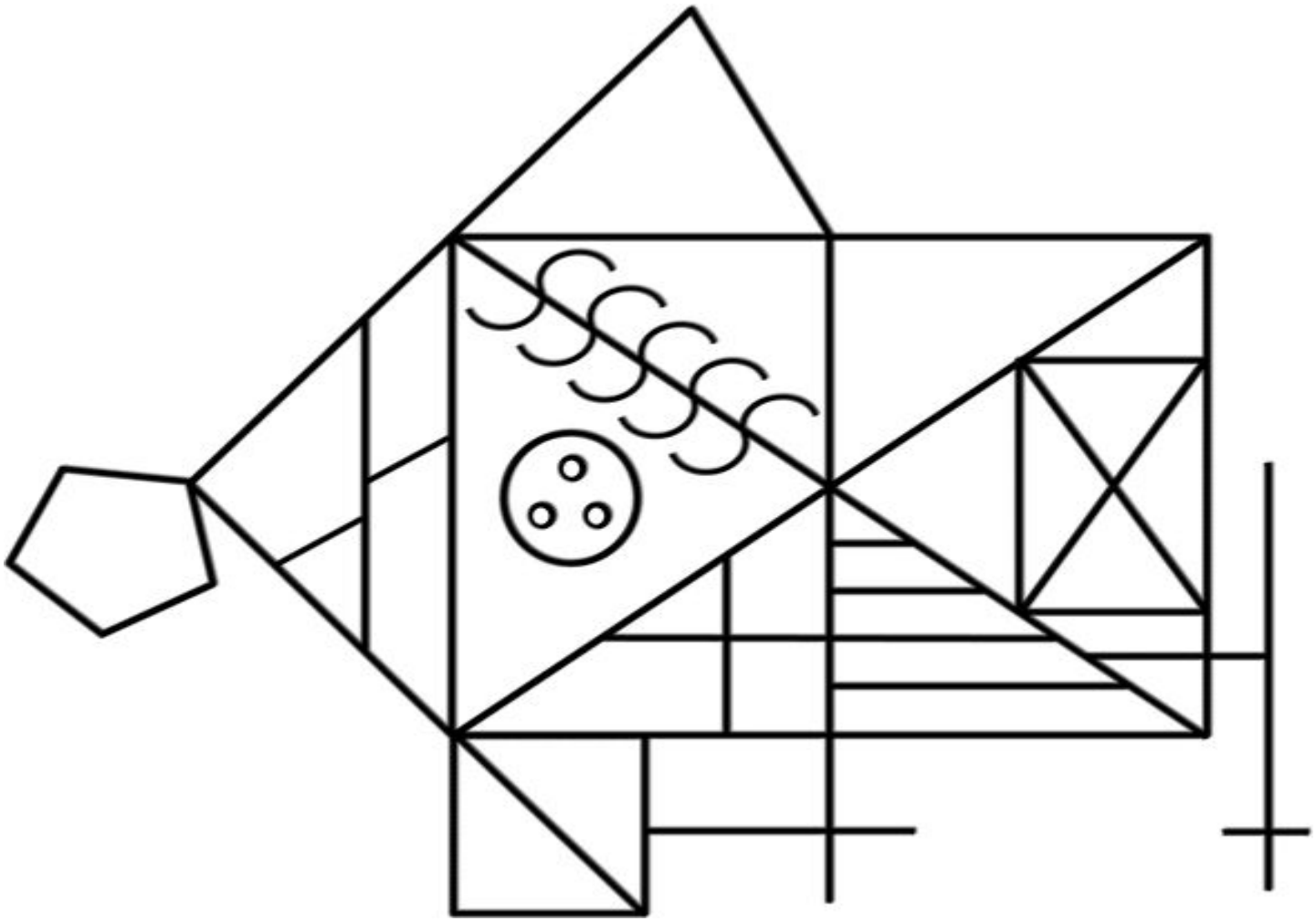
How does metacognition work in the classroom?

- Process questioning
- Generalising
- Challenging
- Teaching rules
- Emphasising order

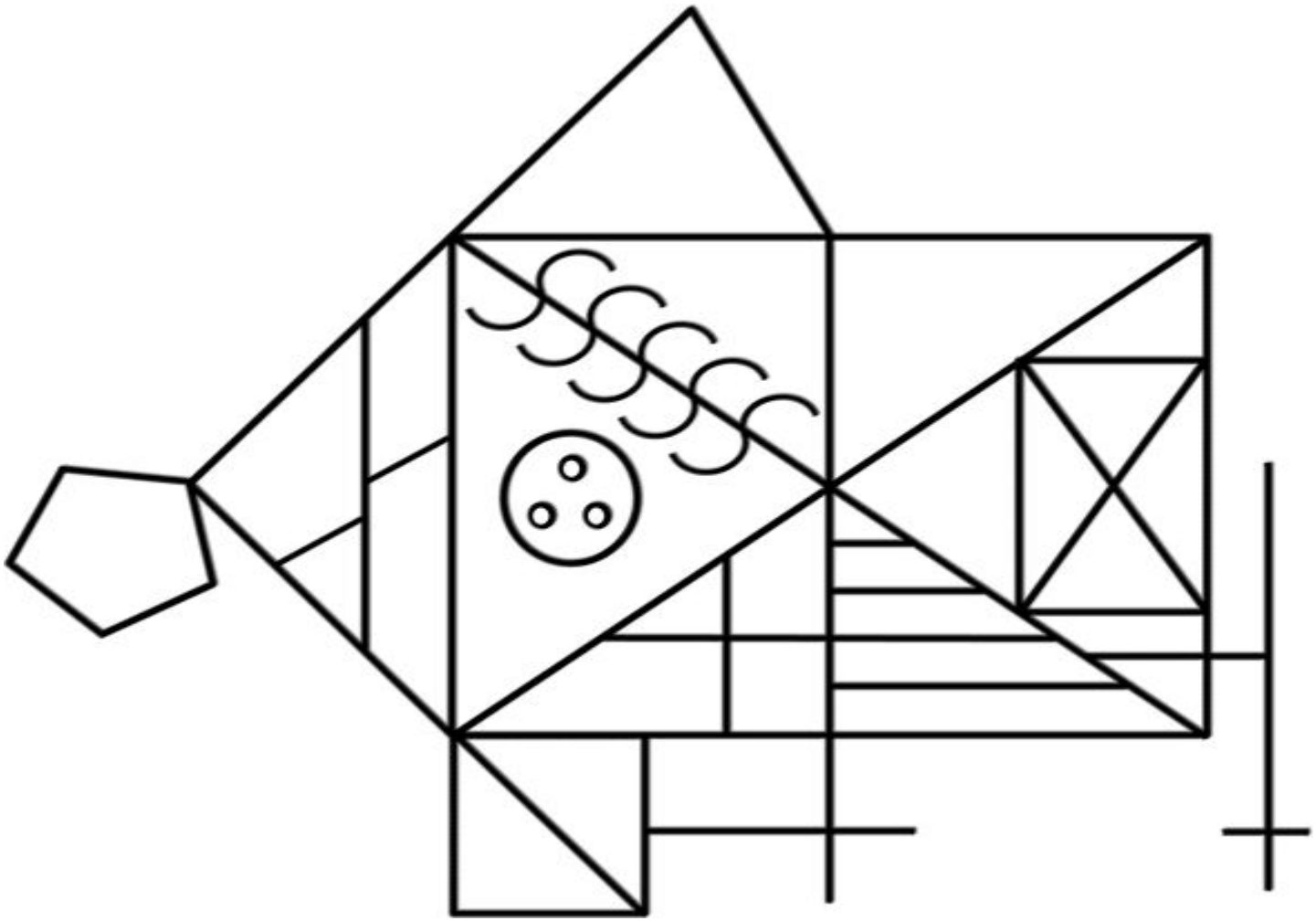
Summary: How do we learn?



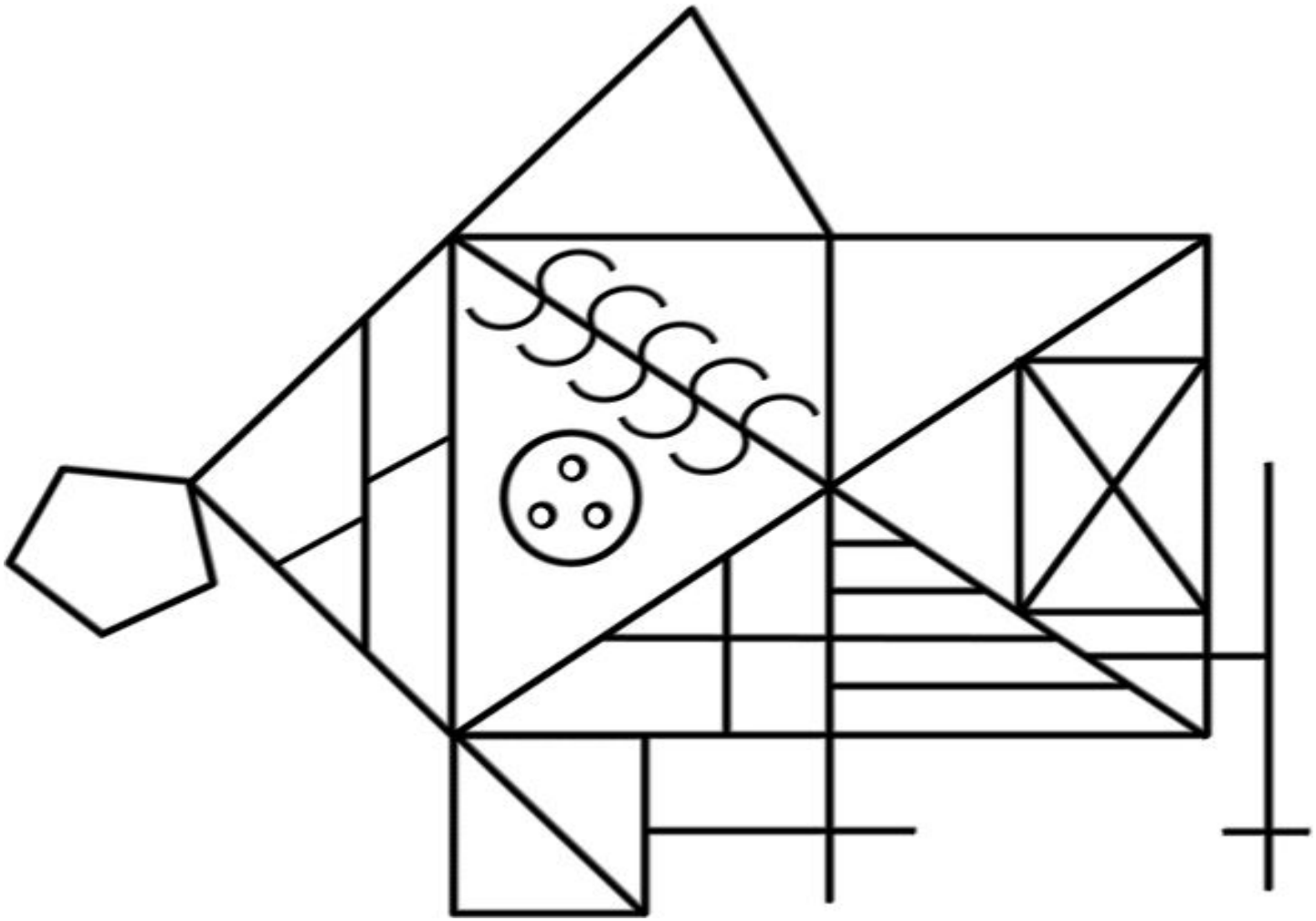
RESEARCH INTO PRACTICE
PRACTICE INTO RESEARCH



Compare



Compare



Review
Plan
Do

Useful questions to make learning ‘visible’

What are you thinking now?

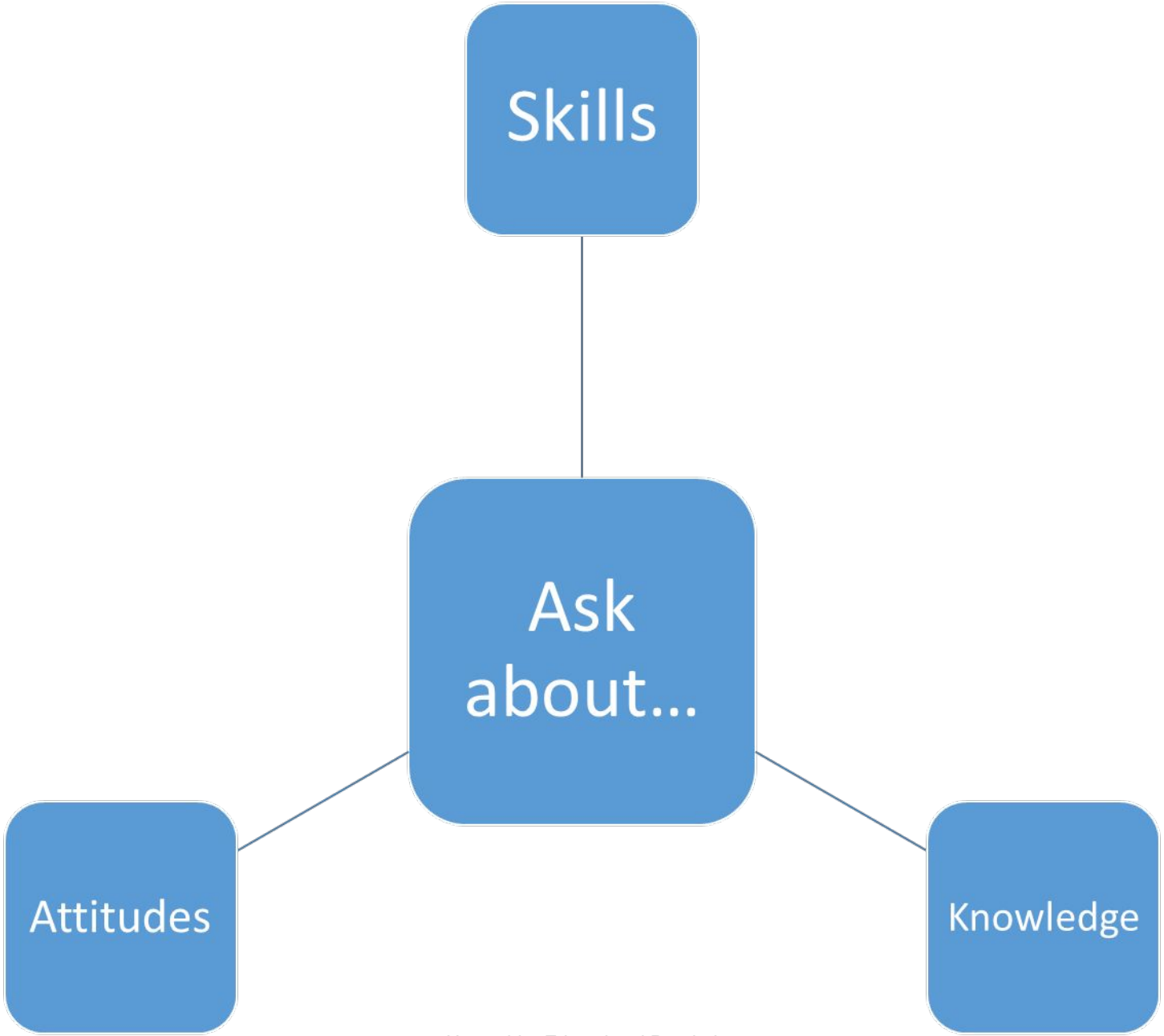
Talk me through what you are doing

Why did you do that?

How do you feel about that?

When would that be useful again?

What could you do next?


Skills
Ask about…
Attitudes
Knowledge

Getting students to **practice** metacognition
Getting adults to **practice** mediation

You have 10 minutes.

Getting students to **practice** metacognition

What did you do?

How did you think your way around the problem?

Did you try to think of something you had done before that might help?

Did you make a plan?

Did you keep checking you were along the right lines?

How did you check you were along the right lines?

Did you talk to yourself or to your group? What did you say?

How would you do it better next time?

When could you use this experience again?

Did you use any knowledge you already had to help you?

How did you feel about the challenge?

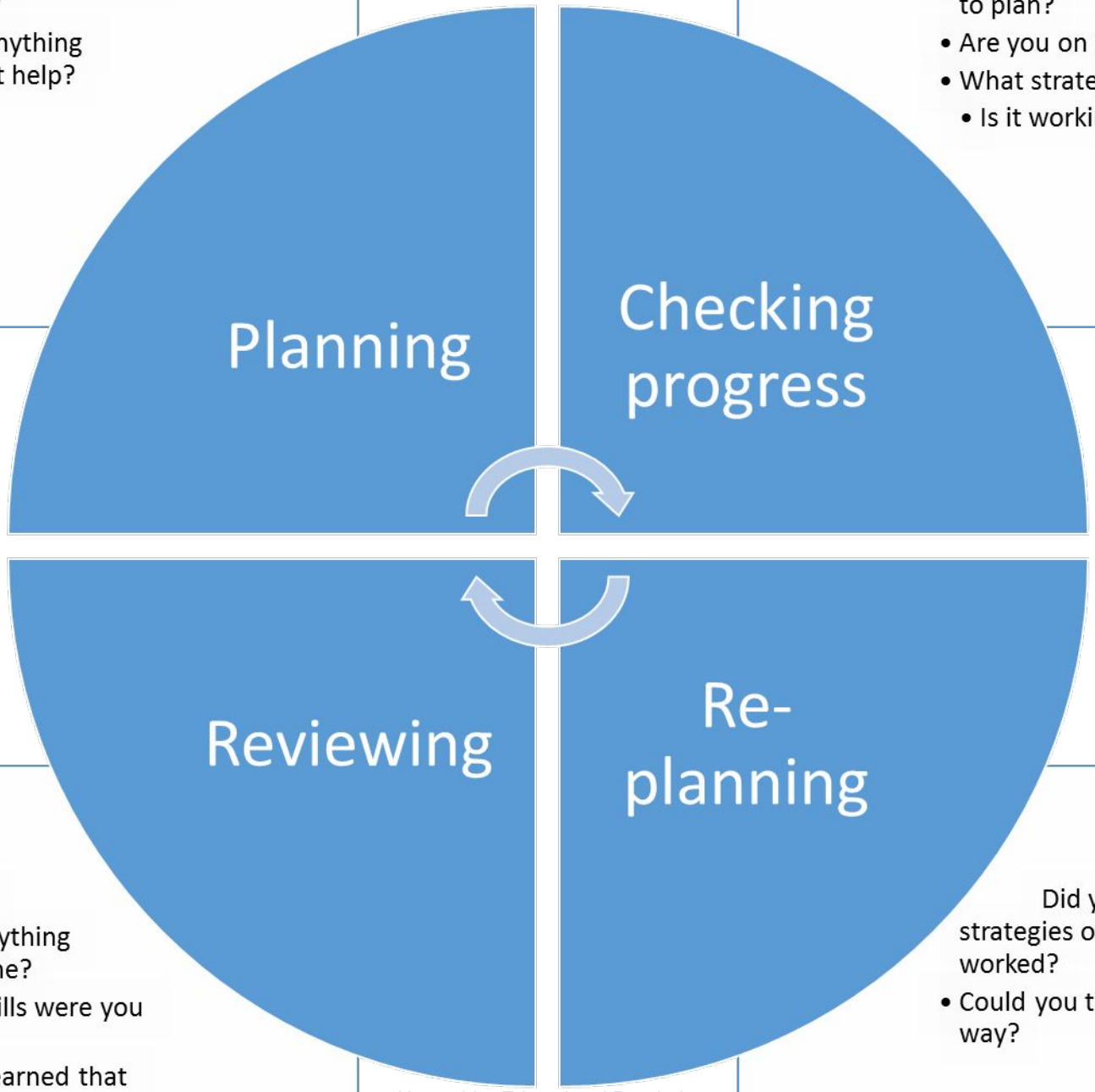
How did you feel about being in a group?

Getting students to **practice** metacognition

Ask ‘guiding’ questions DURING a challenging activity to help make the learning visible and encourage children to use their metacognitive skills.

- What would be a good way of going about this?
- Have you done anything before that might help?

- Is everything going according to plan?
- Are you on track?
- What strategy are you using?
  - Is it working?



- What went well?
- Would you do anything different next time?
- What thinking skills were you using/learning?
- What have you learned that you could use again?

Did you change your strategies or ideas as you worked?

- Could you tackle this another way?

Questions and conclusions